

Quick start for LaTeXing with IEEEtran.cls for IEEE Computer Society Conferences

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Abstract—Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

I. INTRODUCTION

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus enim. Vestibulum pellentesque felis eu massa. **TODO!**

The remainder of the paper starts with a presentation of related work (Section II). It is followed by a presentation of hints on LaTeX (Section III). The paper closes with a conclusion and an outlook on future work (Section IV).

II. RELATED WORK

Winery [1] is a graphical modeling tool. The whole idea of TOSCA is explained by Binz et al. [2].

To search for related work, ScienceOS¹ can be used; it stores data in the EU and backs its answers with citations, which you still have to read and check yourself. Writing hints for student theses are given in the open-access book *Student Thesis Projects*². More hints are given in *Writing for Computer Science*³.

III. LATEX HINTS

This section contains hints on writing LaTeX. It focuses on minimal examples, which can be directly adapted to the content

A. Handling of paragraphs

One sentence per line. This rule is important for the usage of version control systems. A new line is generated with a blank line. As you would do in Word: New paragraphs are generated by pressing enter. In LaTeX, this does not lead to a new paragraph as LaTeX joins subsequent lines.

¹<https://www.scienceos.ai/>

²<https://vdf.ch/product/student-thesis-projects-en.html>

³<https://doi.org/10.1007/978-1-4471-6639-9>

To start a new paragraph, press enter twice. This leads to an empty line. In word, there is the functionality to press shift and enter. This leads to a hard line break. The text starts at the beginning of a new line. In LaTeX, you can do that by using two backslashes (`\`).

Running text seldom needs a hard line break.

Please do *not* use two backslashes for new paragraphs. For instance, this sentence belongs to the same paragraph, whereas the last one started a new one. Andrew Stacey gives a long motivation for that in the section “Line Wrapping” of his *Document Revision System*⁴.

Corresponding LaTeX code of ./paper.tex

```
655 %\begin{document}
656 One sentence per line.
657 This rule is important for the usage of version
    ↪ control systems.
658 A new line is generated with a blank line.
659 As you would do in Word:
660 New paragraphs are generated by pressing enter.
661 In LaTeX, this does not lead to a new paragraph
    ↪ as LaTeX joins subsequent lines.
662 To start a new paragraph, press enter twice.
663 This leads to an empty line.
664 In word, there is the functionality to press
    ↪ shift and enter.
665 This leads to a hard line break.
666 The text starts at the beginning of a new line.
667 In LaTeX, you can do that by using two
    ↪ backslashes (\textbackslash\textbackslash)
    ↪ .
668 \\
669 Running text seldom needs a hard line break.
670
671 Please do \textit{not} use two backslashes for
    ↪ new paragraphs.
672 For instance, this sentence belongs to the same
    ↪ paragraph, whereas the last one started a
    ↪ new one.
673 Andrew Stacey gives a long motivation for that in
    ↪ the section \enquote{Line Wrapping} of
    ↪ his \href{https://loopspace.mathforge.org
    ↪ /HowDidIDoThat/TeX/VCS/#section.3}{\emph{
    ↪ Document Revision System}}\footnote{\url{
    ↪ https://loopspace.mathforge.org/
    ↪ HowDidIDoThat/TeX/VCS/}}.
```

B. Notes separated from the text

The package `mindflow` enables writing down notes and annotations in a way so that they are separated from the main text.

This is a small note.

Corresponding LaTeX code of ./paper.tex

```
680 %\begin{document}
681 \begin{mindflow}
682 This is a small note.
683 \end{mindflow}
```

⁴<https://loopspace.mathforge.org/HowDidIDoThat/TeX/VCS/>

C. Handling TODOs

Marked text.

Corresponding LaTeX code of ./paper.tex

```
688 %\begin{document}
689 \textmarker{Marked text.}
```

With `\textmarker`, only the text color is changed, since this also works well for individual words.

Marked text.

Corresponding LaTeX code of ./paper.tex

```
694 %\begin{document}
695 \textcomment{Marked text.}{A comment on it.}
```

Manual marking for text that has been changed since the last version.

Corresponding LaTeX code of ./paper.tex

```
698 %\begin{document}
699 \modified{Manual marking for text that has been
    ↪ changed since the last version.}
```

This is some text. Changed text.

Corresponding LaTeX code of ./paper.tex

```
702 %\begin{document}
703 This is some text.
704 \change{FL1: text adjusted}{Changed text}.
```

Here is a comment.

Corresponding LaTeX code of ./paper.tex

```
707 %\begin{document}
708 Here is a comment\sidecomment{Comment}.
```

TODO!

Corresponding LaTeX code of ./paper.tex

```
711 %\begin{document}
712 \todo{A lot of text still needs to be produced
    ↪ here}
```

D. Hyphenation

LaTeX automatically hyphenates words. When using microtype, there should be fewer hyphenations than in other settings. You might still need to tweak the hyphenation. Here are some hints:

In case you write “application-specific”, then the word will only be hyphenated at the dash. You can also insert `\allowbreak` into the word, as in `application-specific`, but this is much more effort.

You can now write words containing hyphens which are hyphenated at other places in the word. For that, write " = instead of -, as in application"=specific. This is enabled by an additional configuration of the babel package.

Corresponding L^AT_EX code of ./paper.tex

```

722 æƒüŰőŰ
723 In case you write \enquote{application-specific},
    ↳ then the word will only be hyphenated at
    ↳ the dash.
724 You can also insert \verb1\allowbreak{}1 into the
    ↳ word, as in applica\allowbreak{}tion-
    ↳ specific, but this is much more effort.
725
726 You can now write words containing hyphens which
    ↳ are hyphenated at other places in the
    ↳ word.
727 For that, write \verb1"=1 instead of \verb1-1, as
    ↳ in application"=specific.
728 This is enabled by an additional configuration of
    ↳ the babel package.

```

E. Typesetting Units

Numbers can be written plain text (such as 100), by using the siunitx package as follows: 100 $\frac{\text{km}}{\text{h}}$, or by using plain L^AT_EX (and math mode): 100 $\frac{km}{h}$.

Corresponding L^AT_EX code of ./paper.tex

```

733 æƒüŰőŰ
734 Numbers can be written plain text (such as 100),
    ↳ by using the \href{https://ctan.org/pkg/
    ↳ siunitx}{siunitx} package as follows:
735 \SI{100}{\km\per\hour},
736 or by using plain \LaTeX{} (and math mode):
737 $\textcolor{red}{100}\frac{\mathit{km}}{h}$.
```

5% of 10 kg

Corresponding L^AT_EX code of ./paper.tex

```

740 æƒüŰőŰ
741 \SI{5}{\percent} of \SI{10}{kg}
```

Numbers are automatically grouped: 123 456.

Corresponding L^AT_EX code of ./paper.tex

```

744 æƒüŰőŰ
745 Numbers are automatically grouped: \num{123456}.
```

F. Surrounding Text by Quotes

Please use the “enquote command” to quote something. Quoting with “quote” or “quote” also works.

Corresponding L^AT_EX code of ./paper.tex

```

750 æƒüŰőŰ
751 Please use the \enquote{enquote command} to quote
    ↳ something.
752 Quoting with ``quote'' or ``quote'' also works.
```

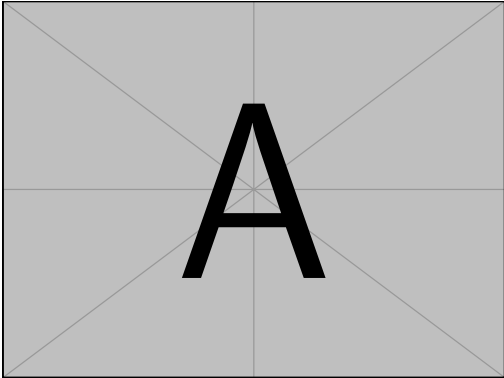


Figure 1. Example figure for zceref demo

Heading1	Heading2
One	Two
Three	Four

Figure 2. Example table for zceref demo

G. Zref-clever examples

zref-clever demonstration: Zceref (short for zceref[S]) at beginning of sentence, zceref in all other cases.

Figure 1 shows a simple fact, although Figure 1 could also show something else.

Figure 2 shows a simple fact, although Figure 2 could also show something else.

Section III-G shows a simple fact, although Section III-G could also show something else.

Corresponding L^AT_EX code of ./paper.tex

```

782 æƒüŰőŰ
783 \Zceref{fig:ex:cref} shows a simple fact, although
    ↳ \zceref{fig:ex:cref} could also show
    ↳ something else.
784
785 \Zceref{tab:ex:cref} shows a simple fact, although
    ↳ \zceref{tab:ex:cref} could also show
    ↳ something else.
786
787 \Zceref{sec:ex:cref} shows a simple fact, although
    ↳ \zceref{sec:ex:cref} could also show
    ↳ something else.
```

H. Figures

Figure 3 shows something interesting.

Corresponding L^AT_EX code of ./paper.tex

```

792 æƒüŰőŰ
793 \Zceref{fig:label} shows something interesting.
794
795 \begin{figure}
796   \centering
797   \includegraphics[width=.8\linewidth]{example-
    ↳ image-golden}
798   \caption[Simple Figure]{
```

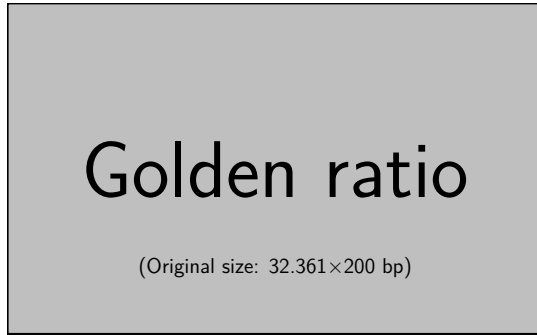


Figure 3. Simple Figure. Based on Scharer [3].



Figure 4. Ulm Minster. Photo: wkrauth, CC0, via Wikimedia Commons.

```

799     Simple Figure.
800     Based on \citet{mwe}.
801   }
802   \label{fig:label}
803 \end{figure}

```

Photos in JPEG or PNG format are included the same way, as shown in Figure 4.

Corresponding L^AT_EX code of ./paper.tex

```

808 %\begin{figure}
809 \centering
810 \includegraphics[width=.5\linewidth]{figures/
811   ↪ ulm-minster.jpg}
812 \caption{Ulm Minster. Photo: wkrauth, CC0, via
813   ↪ Wikimedia Commons.}
813 \label{fig:jpg}
814 \end{figure}

```

One can span a figure across multiple columns by using `\begin{figure*}`. See Figure 5 as an example.

Corresponding L^AT_EX code of ./paper.tex

```

820 %\begin{figure}
821 \begin{figure*}
822 \centering
823 % note that \textwidth is used instead of \
824   ↪ linewidth
824 % This ensures that the graphics width is 60%
825   ↪ of the "page" (text block), and not just
826   ↪ 60% of the current text column
825 % See https://tex.stackexchange.com/a
826   ↪ /17085/9075 for details
826 \includegraphics[width=.6\textwidth]{example-
827   ↪ image-16x9}
827 \caption{16x9 Figure}
828 \label{fig:16x9}
829 \end{figure*}

```

I. Sub Figures

An example of two sub figures is shown in Figure 6.

Corresponding L^AT_EX code of ./paper.tex

```

837 %\begin{figure}
838 \begin{figure*}[!t]
839 \centering
840 \subfloat[Case I]{\includegraphics[width=.4\
841   ↪ linewidth]{example-image-a}%
842   \label{fig:first_case}}
842 \hfil
843 \subfloat[Case II]{\includegraphics[width=.4\
844   ↪ linewidth]{example-image-b}%
845   \label{fig:second_case}}
845 \caption{Example figure with two sub figures.}
846 \label{fig:two_sub_figures}
847 \end{figure*}

```

Note that often IEEE papers with subfigures do not employ subfigure captions (using the optional argument to `\subfloat[]`), but instead will reference/describe all of them (a), (b), etc., within the main caption. Be aware that for `subfig.sty` to generate the (a), (b), etc., subfigure labels, the optional argument to `\subfloat` must be present. If a subcaption is not desired, leave its contents blank, e.g., `\subfloat[]`. An example is shown in Figure 7.

Corresponding L^AT_EX code of ./paper.tex

```

859 %\begin{figure}
860 \begin{figure*}[!t]
861 \centering
862 \subfloat[]{\includegraphics[width=.4\linewidth
863   ↪ ]{example-image-a}%
864   \label{fig:first_case_ieee}}
864 \hfil
865 \subfloat[]{\includegraphics[width=.4\linewidth
866   ↪ ]{example-image-b}%
867   \label{fig:second_case_ieee}}
867 \caption{Example figure with two sub figures.
868   ↪ IEEE style. (a) The first case. (b) The
869   ↪ second case.}
868 \label{fig:two_sub_figures_ieee}
869 \end{figure*}

```

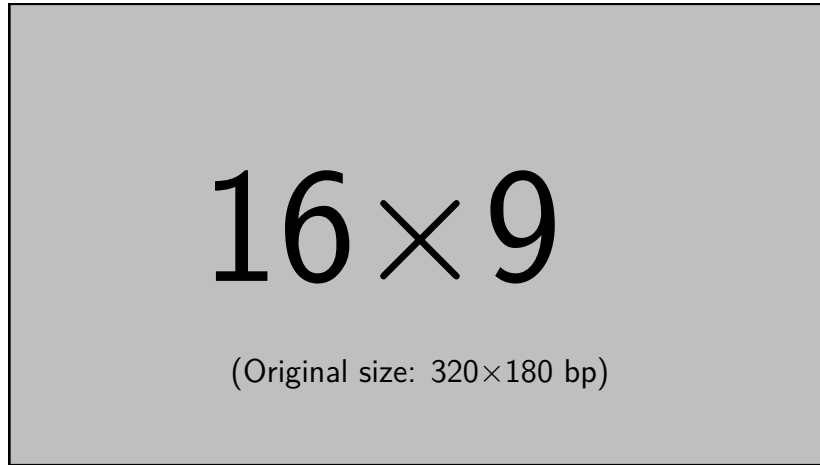
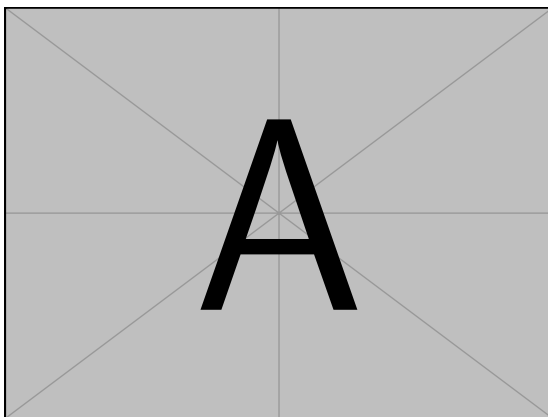
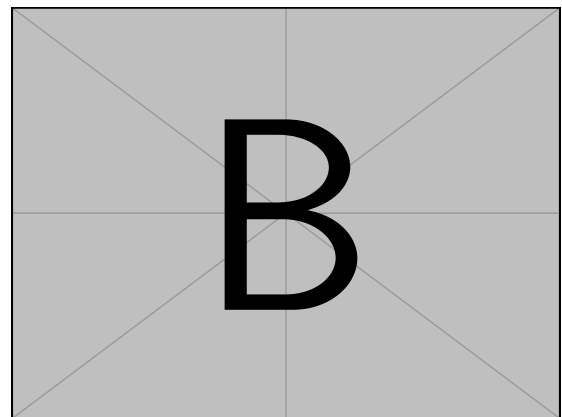


Figure 5. 16x9 Figure

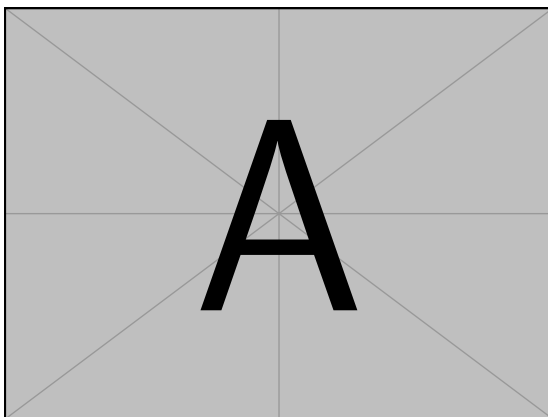


(a) Case I

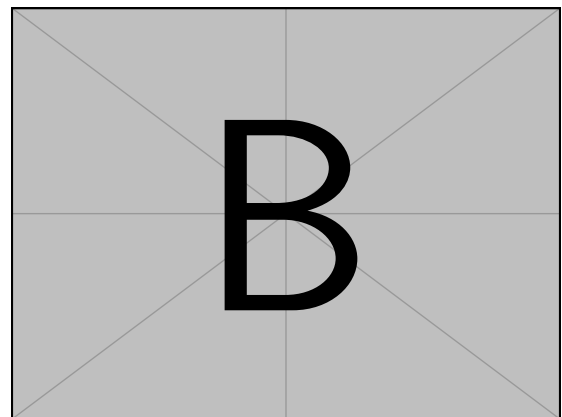


(b) Case II

Figure 6. Example figure with two sub figures.



(a)



(b)

Figure 7. Example figure with two sub figures. IEEE style. (a) The first case. (b) The second case.

J. Figures with tikz

The tikz package creates graphics programmatically. This package generates grids and other regular structures.

Corresponding L^AT_EX code of `./paper.tex`

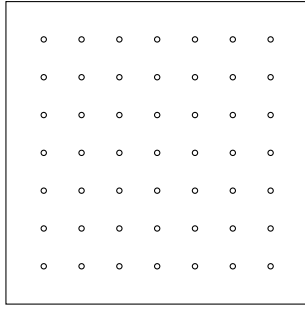


Figure 8. A regular grid generated easily with two for loops.

```

877 %\documentclass{article}
878 \begin{figure}[ht]
879 \centering
880 \begin{tikzpicture}
881 \draw(0,0) rectangle (4,4);
882 \foreach \x in {0.5,1,1.5,2,2.5,3,3.5}
883 \foreach \y in {0.5,1,1.5,2,2.5,3,3.5}
884 \draw(\x,\y) circle (1pt);
885 \end{tikzpicture}
886 \caption[A regular grid generated easily with
887 \hookrightarrow two for loops.]{fig:tikz_example}
888 \end{figure}

```

K. Plots with pgfplots

The pgfplots package plots functions directly in $\text{\LaTeX}$, similar to MATLAB or gnuplot. Some visual examples are available in the package documentation⁵.

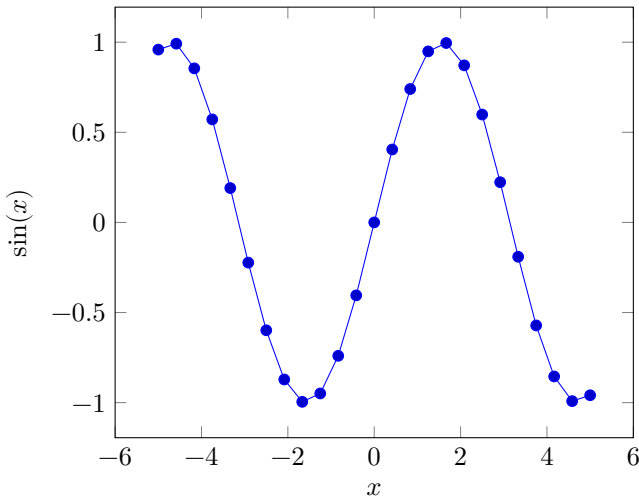


Figure 9. Plot of $\sin(x)$ directly inside the figure environment with pgfplots.

Corresponding $\text{\LaTeX}$ code of ./paper.tex

```

895 %\documentclass{article}
896 \begin{figure}[ht]
897 \centering
898 \begin{tikzpicture}

```

⁵<http://texdoc.net/pkg/pgfplots>

```

899 \begin{axis}[xlabel=$x$, ylabel=$\sin(x)$]
900 \addplot {sin(deg(x))}; % Plot the sine
901 \end{axis}
902 \end{tikzpicture}
903 \caption[Plot of $\sin(x)$ directly inside the
904 \hookrightarrow figure environment with pgfplots.]
905 \label{fig:pgfplots_example}
906 \end{figure}

```

Coordinates can also be given explicitly instead of as a function:

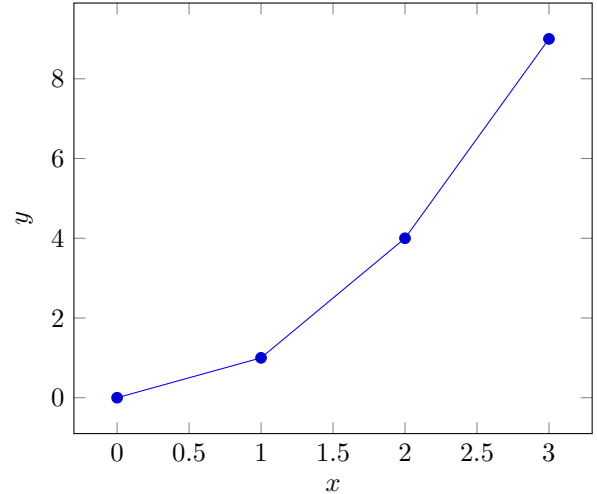


Figure 10. Explicit coordinates plotted with pgfplots.

Corresponding $\text{\LaTeX}$ code of ./paper.tex

```

909 %\documentclass{article}
910 \begin{figure}[ht]
911 \centering
912 \begin{tikzpicture}
913 \begin{axis}[xlabel=$x$, ylabel=$y$]
914 \addplot coordinates {(0,0) (1,1) (2,4) (3,
915 \hookrightarrow 9)}; % Plot explicit coordinates
916 \end{axis}
917 \end{tikzpicture}
918 \caption[Explicit coordinates plotted with
919 \hookrightarrow pgfplots.]
920 \label{fig:pgfplots_coords}
921 \end{figure}

```

L. Tables

Note that IEEE does not support $\text{\LaTeX}$ `\begin{table}`, one has to use $\text{\LaTeX}$ `\begin{figure}`.

Figure 11. Simple Table

Heading1	Heading2
One	Two
Three	Four

Figure 12. Table with diagonal line

Row \ Column	A	B
1	x	y

Corresponding L^AT_EX code of ./paper.tex

```

926 %\begin{figure}
927 \begin{figure}
928 \caption{Simple Table}
929 \label{tab:simple}
930 \centering
931 \begin{tabular}{ll}
932 \toprule
933 Heading1 & Heading2 \\
934 \midrule
935 One & Two \\
936 Three & Four \\
937 \bottomrule
938 \end{tabular}
939 \end{figure}

```

Corresponding L^AT_EX code of ./paper.tex

```

942 %\begin{figure}
943 % Adapted from https://tex.stackexchange.com/a
944 % ↪ /468994/9075
945 \begin{figure}
946 \caption{Table with diagonal line}
947 \label{tab:diag}
948 \begin{center}
949 \begin{tabular}{|l|c|c|}
950 \hline
951 \diagbox{Row}{Column} & A & B \\
952 \hline
953 1 & x & y \\
954 \hline
955 \end{tabular}
956 \end{center}
957 \end{figure}

```

M. Source Code

Listing 1 shows source code written in XML. Line 2 contains a comment.

```

1 <listing name="example">
2 <!-- comment -->
3 <content>not interesting</content>
4 </listing>

```

Listing 1. Example XML Listing

Corresponding L^AT_EX code of ./paper.tex

```

962 %\begin{figure}
963 \Zcref{lst:XML} shows source code written in XML.
964 \Zcref{line:comment} contains a comment.
965
966 \begin{lstlisting}[
967 language=XML,
968 caption={Example XML Listing},
969 label={lst:XML}]
970 <listing name="example">

```

```

1 <listing name="example">
2 Floating
3 </listing>

```

Listing 2. Example XML listing – placed as floating figure

```

1 {
2   key: "value"
3 }

```

Listing 3. Example JSON listing – placed as floating figure

```

971 <!-- comment --> (* \label{line:comment} *)
972 <content>not interesting</content>
973 </listing>
974 \end{lstlisting}

```

One can also add float as parameter to have the listing floating. Listing 2 shows the floating listing.

Corresponding L^AT_EX code of ./paper.tex

```

980 %\begin{figure}
981 \begin{lstlisting}[
982 % one can adjust spacing here if required
983 % aboveskip=2.5\baselineskip,
984 % belowskip=-.8\baselineskip,
985 float,
986 language=XML,
987 caption={Example XML listing -- placed as
988 ↪ floating figure},
989 label={lst:flXML}]
990 <listing name="example">
991 Floating
992 </listing>
993 \end{lstlisting}

```

One can also typeset JSON as shown in Listing 3.

Corresponding L^AT_EX code of ./paper.tex

```

997 %\begin{figure}
998 \begin{lstlisting}[
999 float,
1000 language=json,
1001 caption={Example JSON listing -- placed as
1002 ↪ floating figure},
1003 label={lst:json}]
1004 {
1005   key: "value"
1006 }
1007 \end{lstlisting}

```

Java is also possible as shown in Listing 4.

```

1 public class Hello {
2   public static void main (String[] args) {
3     System.out.println("Hello World!");
4   }
5 }

```

Listing 4. Example Java listing

Corresponding L^AT_EX code of ./paper.tex

```

1011 %\begin{lstlisting}
1012 \begin{lstlisting}[
1013   caption={Example Java listing},
1014   label=lst:java,
1015   language=Java,
1016   float]
1017 public class Hello {
1018     public static void main (String[] args) {
1019         System.out.println("Hello World!");
1020     }
1021 }
1022 \end{lstlisting}

```

N. Itemization

One can list items as follows:

- Item One
- Item Two

Corresponding L^AT_EX code of ./paper.tex

```

1029 %\begin{itemize}
1030 \begin{itemize}
1031   \item Item One
1032   \item Item Two
1033 \end{itemize}

```

With the package paralist, one can create itemizations with lesser spacing:

- Item One
- Item Two

Corresponding L^AT_EX code of ./paper.tex

```

1038 %\begin{compactitem}
1039 \begin{compactitem}
1040   \item Item One
1041   \item Item Two
1042 \end{compactitem}

```

One can enumerate items as follows:

- 1) Item One
- 2) Item Two

Corresponding L^AT_EX code of ./paper.tex

```

1047 %\begin{enumerate}
1048 \begin{enumerate}
1049   \item Item One
1050   \item Item Two
1051 \end{enumerate}

```

With the package paralist, one can create enumerations with lesser spacing:

- 1) Item One
- 2) Item Two

Corresponding L^AT_EX code of ./paper.tex

```

1056 %\begin{compactenum}
1057 \begin{compactenum}
1058   \item Item One
1059   \item Item Two
1060 \end{compactenum}

```

With paralist, one can even have all items typeset after each other and have them clean in the TeX document:

1) All these items... 2) ...appear in one line 3) This is enabled by the paralist package.

Corresponding L^AT_EX code of ./paper.tex

```

1065 %\begin{inparaenum}
1066 \begin{inparaenum}
1067   \item All these items...
1068   \item ...appear in one line
1069   \item This is enabled by the paralist package.
1070 \end{inparaenum}

```

O. Other Features

The words “workflow” and “dwarflike” can be copied from the PDF and pasted to a text file.

Corresponding L^AT_EX code of ./paper.tex

```

1075 %\enquote{workflow} and \enquote{dwarflike}
1076 \enquote{The words \enquote{workflow} and \enquote{dwarflike} can be copied from the PDF and
        \enquote{pasted to a text file.}

```

The symbol for powerset is now correct: $\wp$ and not a Weierstrass p ($\wp$).

$\wp(1,2,3)$

Corresponding L^AT_EX code of ./paper.tex

```

1079 %\powerset{{1,2,3}}
1080 \powerset{The symbol for powerset is now correct: $\wp$ and not a Weierstrass p ($\wp$)}
1081 \powerset{.}
1082 \powerset{{1,2,3}}

```

Brackets work as designed: <test> One can also input backticks in verbatim text: ``test``.

Corresponding L^AT_EX code of ./paper.tex

```

1085 %\brackets{test}
1086 \brackets{Brackets work as designed: <test>}
1087 \brackets{One can also input backticks in verbatim text: \verb`test`.}

```

Acronyms can be typeset in running text with `\initialism`, which sets them as small caps that blend into the surrounding lowercase text. It is the lightweight alternative to the full acronym and glossary management

(\gls, with an automatic list of abbreviations) provided by the thesis variants: \initialism only typesets the acronym and does not add it to any list. The macro and the ready-made acronyms (\OMG, \BPEL, \BPMN, \UML) are defined in `commands.tex`, where you can add your own.

The UML is maintained by the OMG; related standards include BPEL and BPMN. Any acronym can be set inline with REST or HTTP.

Corresponding L^AT_EX code of ./paper.tex

```

1093 æŒüŦŦŦ
1094 The \UML is maintained by the \OMG; related
      ↳ standards include \BPEL and \BPMN.
1095 Any acronym can be set inline with \initialism{
      ↳ REST} or \initialism{HTTP}.

```

IV. CONCLUSION AND OUTLOOK

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

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ACKNOWLEDGMENT

Identification of funding sources and other support, and thanks to individuals and groups that assisted in the research and the preparation of the work should be included in an acknowledgment section, which comes right before the reference section in your document [4].

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All links were last followed on October 5, 2020.